torch.std#

https://pytorch.org/docs/stable/generated/torch.std.html

Description:

Calculates the standard deviation over the dimensions specified by dim. dim can be a single dimension, list of dimensions, or None to reduce over all dimensions. The standard deviation (σ\sigmaσ) is calculated as where xxx is the sample set of elements, x̄\bar{x}x̄ is the sample mean, NNN is the number of samples and δN\delta NδN is the correction. If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s). input (Tensor) – the input tensor.

Function Signature

```
torch.std(input, dim=None, *, correction=1, keepdim=False,
out=None) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

correction (int) – difference between the sample size and sample degrees of freedom. Defaults to Bessel's correction, correction=1. Changed in version 2.0: Previously this argument was called unbiased and was a boolean with True corresponding to correction=1 and False being correction=0. keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False. out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.tensor(
...     [[ 0.2035,  1.2959,  1.8101, -0.4644],
...      [ 1.5027, -0.3270,  0.5905,  0.6538],
...      [-1.5745,  1.3330, -0.5596, -0.6548],
...      [ 0.1264, -0.5080,  1.6420,  0.1992]]
... )  # fmt: skip
>>> torch.std(a, dim=1, keepdim=True)
tensor([[1.0311],
        [0.7477],
        [1.2204],
        [0.9087]])
```

Detailed Documentation

Documentation

Calculates the standard deviation over the dimensions specified by dim. dim can be a single dimension, list of dimensions, or None to reduce over all dimensions.

The standard deviation (σ\sigmaσ) is calculated as

where xxx is the sample set of elements, x¯\bar{x}x¯ is the sample mean, NNN is the number of samples and δN\delta NδN is the correction.

If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s).

input (Tensor) – the input tensor.

dim (int or tuple of ints, optional) – the dimension or dimensions to reduce. If None, all dimensions are reduced.

difference between the sample size and sample degrees of freedom. Defaults to Bessel’s correction, correction=1.

Changed in version 2.0: Previously this argument was called unbiased and was a boolean with True corresponding to correction=1 and False being correction=0.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (Tensor, optional) – the output tensor.